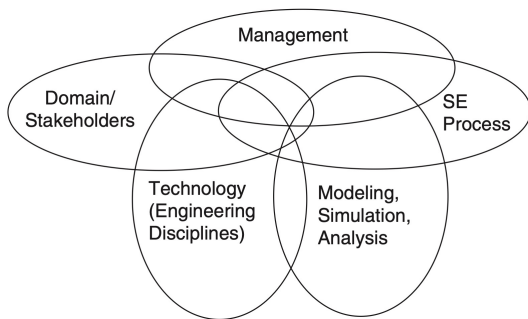


Project Planning Process & Project Charter Components

DAEN 459 Fall 2025
Alexander Abuabara



Previously ... team formation



Expertise required on the engineering team for a system.

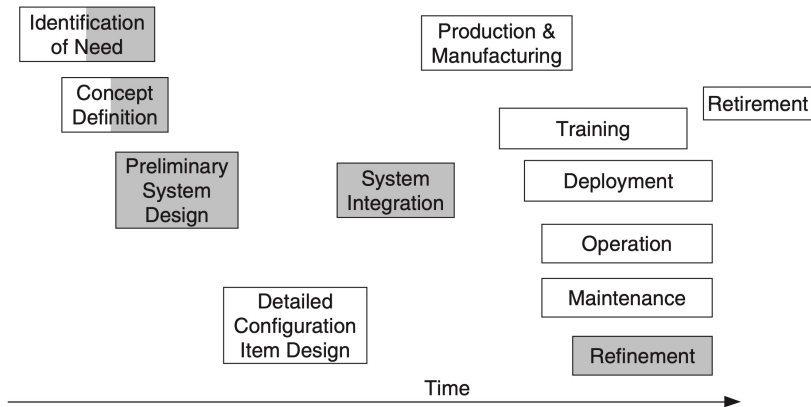
Project Planning & the Project Charter

- **Gather information:** Consolidate meeting notes and stakeholder needs.
- **Clarify objectives:** Scope, goals, business case.
- **Stakeholder alignment:** Roles, responsibilities, approvals.
- **Project structure:** Deliverables, milestones, timeline overview.
- **Risk awareness:** Key assumptions, constraints, and potential risks.

Project Planning Process

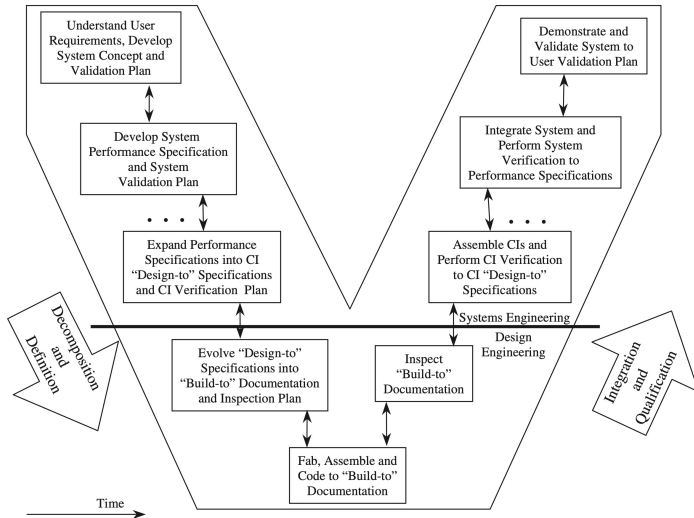
- Define scope, objectives, and deliverables
- Identify stakeholders and gather requirements
- Develop the Work Breakdown Structure (WBS)
(draft, to be refined through iteration)
- Estimate resources, time, and costs
- Sequence activities and create a project schedule
- Identify risks and plan mitigations
- Finalize project charter (project management plan)

Phases of the System Life Cycle



One of many possible representations of the life cycle of a system.

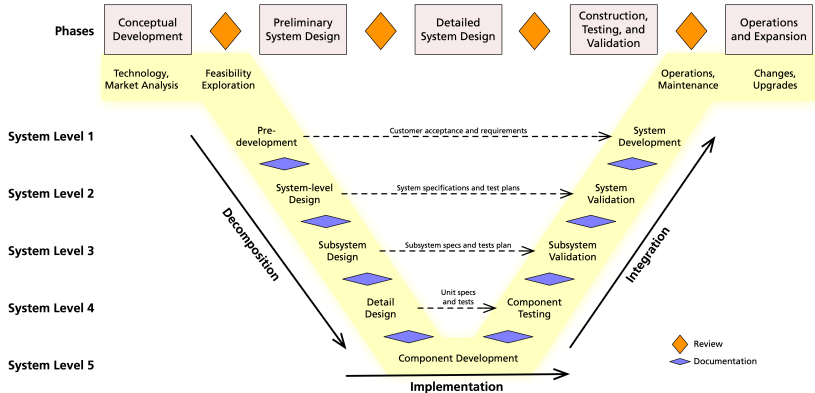
V-model



Systems engineering "Vee" after Forsberg and Mooz [1992].

(Extracted from "The engineering design of systems - models and methods", p.10.)

V-model



Ideally we should aim one level for every 1-2 weeks.
(Prepared by the author, adapted for use in this course.)

Project Charter Components

- **Purpose & Objectives:** why the project exists
- **Scope Statement:** boundaries of work
- **High-Level Requirements:** essential deliverables
- **Stakeholders & Roles:** sponsor, team, clients
- **Milestones & Timeline:** major phases and deadlines
- **Budget Overview:** resource allocation, all kinds (e.g., skills, costs, equipment, and time)
- **Risks & Assumptions:** known challenges, internal/external threats
- **Validation & Verification:** how to meet intended purposes
- **Approval & Sign-Off:** formal authorization

Validation & Verification (V&V)

- **Validation:** are we building the *right* solution?
Ensure alignment with sponsor needs and project objectives.
Review requirements and constraints early.
- **Verification:** are we building the solution *right*?
Test deliverables against defined requirements.
Iterative reviews, simulations, and peer evaluations.
- Continuous process throughout **project lifecycle**.

Goal for This Week: Find an Advisor

- **Primary advisor (required):** ISEN faculty member.
(I can recommend matches once I review your initial notes)
- **Role:** subject-matter expert; offers guidance, feedback, and technical direction.
- **Cadence:** meet about every two weeks unless an alternative schedule is approved.
- **Professionalism:** treat advisors as part of the team: communicate clearly and respectfully. (ensure a formal, businesslike approach at all times).
- **Secondary advisor?** (optional): from TAMU or even external (needs approval with brief justification).

Which should we focus on?

